```
> c("One", "Two", "Three", "Four")
[1] "One"   "Two"   "Three" "Four"
```

You can retrieve members by specifying an index or an index range for the vector. A couple of examples are given below:

```
> v <- c("One", "Two", "Three", "Four")

> v[1]
[1] "One"

> v[1:3]
[1] "One"   "Two"   "Three"
```

In the above example, the vector represents a character vector.

```
> typeof(v)
[1] "character"
```

You can also use logical values in a vector. For example:

```
> b <- c(TRUE, FALSE)

> b
[1]  TRUE FALSE

> typeof(b)
[1] "logical"
```

The R *ifelse* is a conditional element selection function. An example is given below to illustrate its use:

```
> p <- c("0", "0", "1", "1")

> q <- c("0", "1", "0", "1")

> ifelse(c(FALSE, FALSE, TRUE, TRUE), p, q)
[1] "0" "1" "1" "1"
```

## Array

An array is a multi-dimensional vector where you can explicitly define the dimensions. For example:

```
> a <- array(c(1, 2, 3, 4, 5, 6, 7, 8, 9), dim=c(3, 3))
> a
     [,1] [,2] [,3]
[1,]    1    4    7
[2,]    2    5    8
[3,]    3    6    9
```

You can retrieve a member by specifying the indices within a square parenthesis:

```
> a[2,3]
[1] 8
```

You can also mention a range of indices. For example:

```
> a[1:2, 1:2]
     [,1] [,2]
[1,]    1    4
[2,]    2    5
```

The following construct returns all the values for the first row:

```
> a[1,]
[1] 1 4 7
```

You can specify the column number to return all its values as well:

```
> a[,2]
[1] 4 5 6
```

The output can be delimited by specifying an index range, as shown below:

```
> a[2:1, ]
     [,1] [,2] [,3]
[1,]    2    5    8
[2,]    1    4    7
```

The following is an example of a 2x2x2 multi-dimensional array construction:

```
> m <- array(c(1, 2, 3, 4, 5, 6, 7, 8), dim=c(2, 2, 2))
> m
, , 1

     [,1] [,2]
[1,]    1    3
[2,]    2    4

, , 2

     [,1] [,2]
[1,]    5    7
[2,]    6    8
```

You can again specify the indices for the various dimensions to retrieve the values from the defined array: